

do63_clustering_advanced-views

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Soft, Model-Based, and Hierarchical Views

Clustering beyond hard partitions: memberships instead of single labels, probabilistic mixtures instead of geometric distance, dendrograms instead of flat groups, and graph communities instead of a feature matrix at all.

4 TECHNIQUES • CLUSTER_CMEANS • CLUSTER_GMM • CLUSTER_HCLUST • CLUSTER_LOUVAIN_GRAPH

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Widening the Clustering Perspective

`clu_kmeans` and `clu_pam` (partitional) and `clu_dbscan` (density-based) all share one assumption: every instance belongs to exactly one cluster, decided by geometric proximity in a feature matrix. This deck breaks that assumption in four different directions.

- **Soft membership** — an instance can partially belong to several clusters at once.

- **Probabilistic structure** — clusters are modeled as overlapping statistical distributions, not hard regions.
- **Hierarchy** — clusters are nested at multiple resolutions, not fixed at one `k`.
- **Graphs, not points** — the input is a network of relationships, not a table of coordinates.

Fuzzy Membership — `cluster_cmeans`

Fuzzy c-means clustering keeps membership degrees instead of only hard assignments — every instance gets a probability-like score for every cluster, not just one final label.

```
model <- cluster_cmeans(centers = 3, m = 2)
model <- fit(model, x)
clu <- cluster(model, x)
head(attr(clu, "membership"))
```

- `m` controls fuzziness: higher values produce softer, more overlapping memberships.
- The membership matrix is attached to the clustering result — useful when a hard yes/no cluster call would hide genuinely ambiguous points.

Reference: Bezdek, J. C. (1981). *Pattern Recognition with Fuzzy Objective Function Algorithms*.

Probabilistic Mixtures — `cluster_gmm`

Gaussian Mixture Model clustering models the data as a mixture of overlapping Gaussian distributions rather than partitioning it by distance to a centroid or medoid.

```
model <- cluster_gmm(G = 3)
model <- fit(model, x)
clu <- cluster(model, x)
```

- `G` sets the number of mixture components (analogous to `k` in k-means).
- Clusters can be elliptical and overlapping — a natural fit when partitional methods' spherical assumption does not hold.

Reference: Fraley, C., and Raftery, A. E. (2002). *Model-Based Clustering, Discriminant Analysis, and Density Estimation*.

Hierarchical Structure — `cluster_hclust`

Agglomerative hierarchical clustering builds a full merge tree that can be inspected before or after cutting it into a fixed number of clusters — the dendrogram itself is part of the deliverable, not just the final labels.

```
model <- cluster_hclust(k = 3, method = "ward.D2")
model <- fit(model, x)
clu <- cluster(model, x)
plot_dendrogram(model$hc, title = "Hierarchical clustering of iris")
```

- `method = "ward.D2"` minimizes within-cluster variance at each merge step.
- The dendrogram lets the reader choose a different cut (`k`) after the fact, without refitting.

Reference: Johnson, S. C. (1967). *Hierarchical Clustering Schemes*.

Graph Communities — `cluster_louvain_graph`

The one deliberate exception in the whole clustering family: the input is a graph, not a feature matrix, and the output is communities defined by modularity optimization, not geometric groups.

```
g <- igraph::sample_gnp(n = 20, p = 0.15)
model <- cluster_louvain_graph()
model <- fit(model, g)
clu <- cluster(model, g)
attr(clu, "modularity")
```

- No `iris[, 1:4]` -style feature matrix — the model consumes an `igraph` object directly.
- **Common mistake:** comparing this too literally against centroid-based clustering, as if both solved the same geometric problem — they don't.
- **Common mistake:** expecting a reference label like `Species` for evaluation; many graph problems are fully unsupervised with no external ground truth.
- Modularity is a structural quality indicator for *this* partition on *this* graph — not an absolute guarantee of "true" communities.

Reference: Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). *Fast Unfolding of Communities in Large Networks*.

What to Observe

- The `daltoolbox` rhythm — configure, fit, cluster, evaluate — survives even the graph exception; only the input type and the interpretation change.
- Choosing among these four is a choice about what "belonging to a cluster" should mean: fuzzy, probabilistic, hierarchical, or relational.